GEOSPATIAL PORTFOLIO 2024

Muhammad Faizal Rafli

Geospatial Analyst & Remote Sensing Specialist

GEE · InSAR · WebGIS · Python · PostGIS · Volcanic Hazard

S2 · ITB · Geodesi & Geomatika · 2024–2026

S1 · UNPAD · Geofisika · GPA 3.55

BRIN Research Assistant · UNPAD · ITB

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Bandung, Indonesia

HaK Cipta
CitarumView

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01

PROFIL & KOMPETENSI

Muhammad Faizal Rafli

Geospatial Analyst
Remote Sensing Specialist

3.55	3.51	2+ Thn
IPK S1	IPK S2	BRIN

Keahlian Teknis

GIS & Remote Sensing

ArcGIS · QGIS · Google Earth Engine (GEE) · ENVI · SNAP

InSAR & SAR

Interferometric SAR (InSAR) · Deformasi Permukaan · Sentinel-1

Programming

Python · SQL · PostGIS · JavaScript

Web GIS Development

Google Earth Engine Apps · WebGIS Development · GEE API

Domain Expertise

Volcanic Hazard · Urban Heat Island · Environmental Geoscience

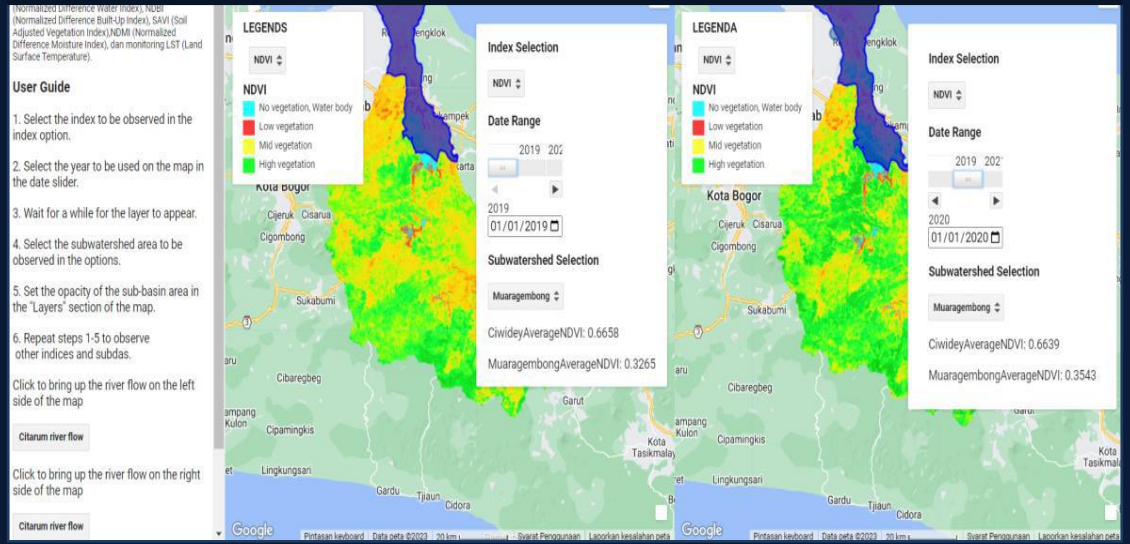
Bahasa: Indonesia | English (Professional)

02

CitarumView

Web GIS Monitoring Sungai Citarum Berbasis Google Earth Engine

code.earthengine.google.com/CitarumView



Platform

Google Earth Engine (GEE)

Institusi

Universitas Padjadjaran

Tahun

2022–2023

Status

Hak Cipta No. 2023

Publikasi Ilmiah

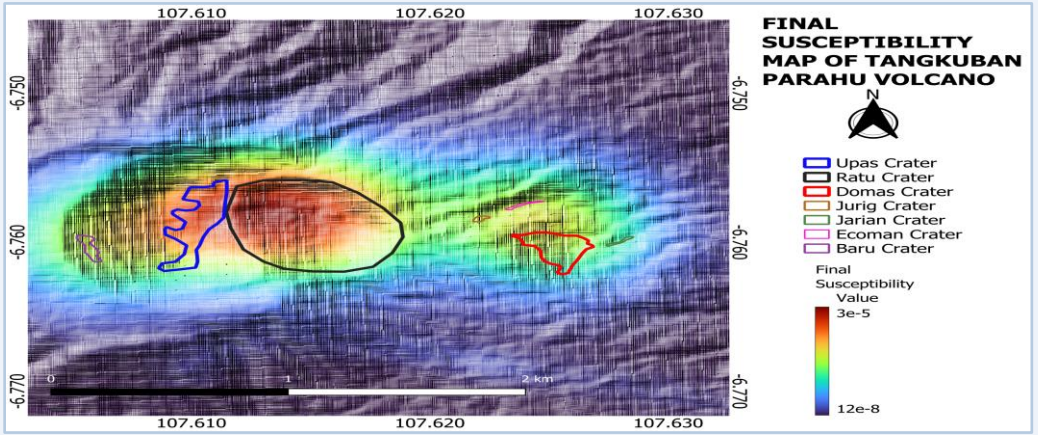
IOP Conf. Ser: Earth Environ. Sci.
1266 (2023) 012011

Fitur Utama:

- Monitoring kualitas air
- Analisis temporal
- Visualisasi interaktif
- Berbasis GEE

PEMETAAN BAHAYA VULKANIK

Volcanic Susceptibility Mapping Gunung Tangkubanparahu



Institusi: BRIN (National Research & Innovation Agency)

Periode: September 2023 – Oktober 2025 · Lokasi: Jawa Barat

Pemetaan susceptibility vulkanik jangka panjang menggunakan metode analisis spasial multi-parameter untuk identifikasi zona bahaya dan mendukung manajemen risiko bencana gunung api.

Metodologi & Tools

Data Input

DEM, citra satelit, data seismik, inventaris letusan

Analisis Spasial

Weighted overlay, susceptibility modelling, GIS analysis

Platform

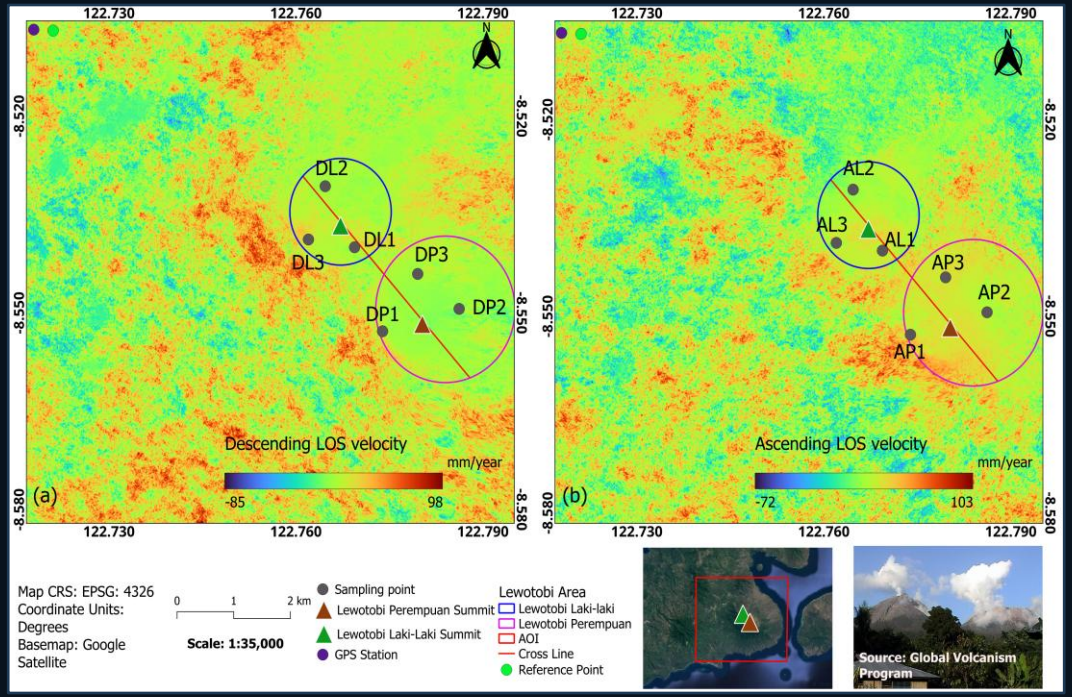
Google Earth Engine, QGIS, ArcGIS, Python

Output

Peta zonasi bahaya vulkanik multi-skenario

Analisis Deformasi InSAR

Gunung Lewotobi · Interferometric Synthetic Aperture Radar · BRIN



Sensor

Sentinel-1 SAR (C-band)

Metode

SBAS InSAR

Software

GAMMA SAR, Python

Parameter

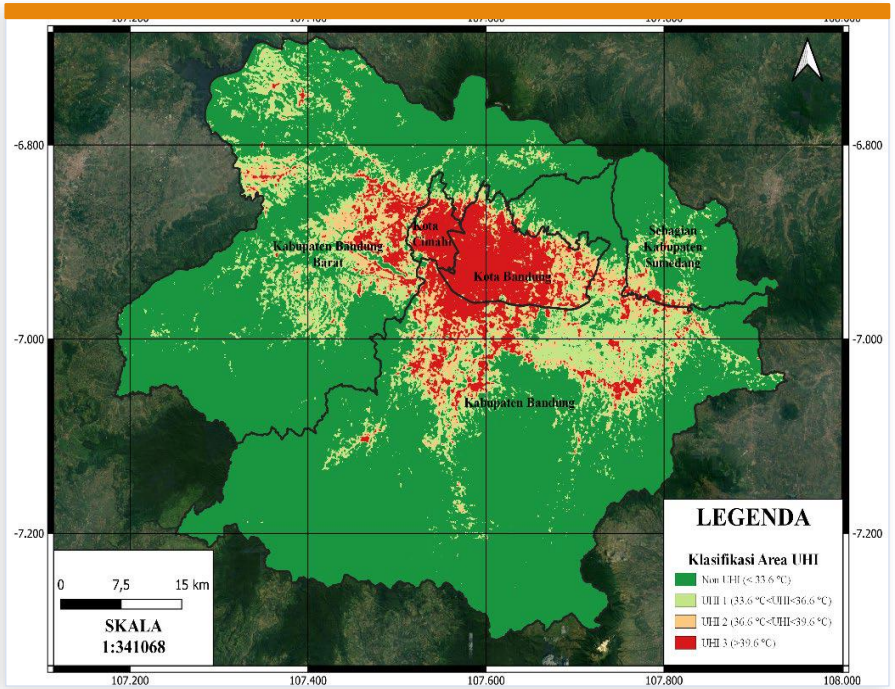
Line-of-Sight Displacement, Fringe Pattern

Tujuan

Monitoring aktivitas vulkanik & deteksi deformasi pra-erupsi

BRIN Research · 2023–2025

URBAN HEAT ISLAND — CEKUNGAN BANDUNG



Skripsi S1 · UNPAD · 2023–2024

Detail Analisis

Land Surface Temperature (LST)

Retrieval suhu permukaan lahan dari citra termal Landsat-8 menggunakan algoritma split-window.

Urban Heat Island Intensity

Perhitungan intensitas UHI antara kawasan urban dan rural di wilayah Cekungan Bandung.

Korelasi Tutupan Lahan

Analisis hubungan antara NDVI, NDBI, dan distribusi suhu permukaan lahan.

Temporal Analysis

Perbandingan multi-temporal untuk melihat tren perubahan UHI seiring perkembangan kota.

Academic Writing

Volcanic Eruption Susceptibility Analysis of Tangkuban Parahu Volcano, West Java, Indonesia

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[3] Faculty of Earth Sciences and Technology, Bandung Institute of Technology (ITB), Jl. Ganesha No. 10, Bandung, West Java 40132, Indonesia

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ABSTRACT

Volcanic eruptions are dynamic events with the potential to occur at multiple changing locations, making continuous monitoring essential for predicting future eruption sites. This study presents a multi-criteria assessment to evaluate long-term spatial susceptibility at Tangkuban Parahu Volcano in Indonesia's Sunda Volcano Complex. Using a spatial analysis approach and geological data, we generated susceptibility maps predicting potential eruption sites. The analysis integrated geological databases incorporating volcanic and tectonic features at Tangkuban Parahu. Results highlight the importance of historical geological data, particularly from the Ratu, Upas, and Domas craters, in modeling eruption scenarios. The maps indicate areas with the highest likelihood of future vent openings near these key summit craters. The long-term susceptibility map developed in this study provides a critical foundation for creating short-term susceptibility maps for future eruption scenarios. Future studies can build on this map by incorporating near-real-time data to enhance volcanic hazard mitigation.

Keywords: Susceptibility, Tangkuban Parahu, Volcanic hazard, Eruptive vent, Volcano-tectonics, Spatial clustering.

Unveiling Subsurface Volume Dynamics: Spatio-Temporal Deformation Analysis of Lewotobi Volcano using SBAS-InSAR Time Series and Inversion via Mogi Source

Muhammad Faizal Rafli[a,c*], Irwan Gumilar[b,], Estu Kriswati[c], Teguh Purnama Sidiq[b]

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[b] Geodetic Science, Engineering and Innovation Research Group, Faculty of Earth Sciences and Technology, Institut Teknologi Bandung, Indonesia

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ABSTRACT

Lewotobi is one of the active volcanoes in Eastern Indonesia. It is a complex stratovolcano with a long history of eruptions, but a deeper understanding of its deformation dynamics is still limited. Long-term and precise surface deformation monitoring plays an important role in understanding the dynamics of a volcano prior to an eruption. This study aims to monitor the surface deformation dynamics of Lewotobi over an 8-year period and determine the depth, location, and volume changes of magma source under two peaks using geodetic inversion.. This study uses the Small Baseline Subset InSAR (SBAS-InSAR) technique on Sentinel-1 satellite data to map and calculate the spatio-temporal rate of surface deformation in the Lewotobi area during the period 2018 to 2025. The resulting deformation data will then be interpreted using Mogi volume source modeling to characterize the source of deformation. The deformation investigation results identified inflation occurring at both peaks with the deformation rates of Lewotobi Laki-laki is 34.3±1.8 mm/year and Lewotobi Perempuan is 20.3±1.7 mm/year. Mogi modeling shows that the accumulated volume change for Lewotobi is $2.32\times10^6 m^3$ at a depth of 1357 meters . This shallow source depth indicates relatively rapid magma migration and a high potential for danger, requiring vigilance in monitoring. The Mogi modeling results obtained in this study are a form of first-order estimation or initial characterization to determine preliminary information about the location of the source point as material for consideration in further research.

Keywords: Lesser sunda islands, Satellite geodesy, Source modelling, Shallow source

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IOP Conf. Series: Earth and Environmental Science 1266 (2023) 012011 doi:10.1088/1755-1315/1266/1/012011

CitarumView: A Google Earth Engine–based application for Citarum River monitoring

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Abstract. The Citarum River is a vital waterway in West Java, Indonesia, that sustains life along its sub-watershed. However, the river is facing significant environmental challenges, such as deforestation, and land-use changes, which have led to a decline in its ecosystem quality. To address this issue, the study aimed to develop an application called "CitarumView" that uses remote sensing methods to assess changes in index values focused on the condition of vegetation, built-up, soil moisture, water body, and surface temperature and obtain a description of the Citarum watershed condition in 2019 and 2020. The Google Earth Engine was used to collect and process satellite imagery data, including visual bands B4 (red), B3 (green), B2 (blue), Near-infrared (NIR), thermal infrared (B10 Landsat-8), and short-wave infrared (SWIR), from Sentinel-2 and Landsat-8 satellites. The CitarumView application provides a tool for decision-makers, researchers, and the general public to continuously monitor the river's condition. The application also provides information on the changes in the ecosystem quality over time, allowing for more informed decisions to be made concerning river management. The analysis conducted using CitarumView revealed that the average of vegetation index increased by 6% with low to mid vegetation level, the max of water index increased by 4.5%, the average of moisture index increased by 39% with mid canopy water, the average of built-up index decreased by 28%, the average of soil adjusted vegetation index increased by 12% with mid to high vegetation, and the average of surface temperature decreased by 5% for 2019–2020.

Pencapaian & Kontak

Hak Cipta Program
2023

CitarumView — GEE-based
Citarum River Monitoring App

Publikasi Ilmiah
2023

IOP Conf. Ser. Earth Environ. Sci.
DOI: 10.1088/1755-1315/1266/1/012011

Presenter Internasional
2023

Int'l Seminar Natural Resources
& Environmental Mgmt. · IPB · Bogor

Pengalaman Kerja

Research Assistant · BRIN

Susceptibility mapping, InSAR deformasi Lewotobi

Peneliti · UNPAD

CitarumView GEE App, analisis Urban Heat Island Cekungan Bandung

Mari Berkolaborasi!

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